INDIAN INSTITUTE OF TECHNOLOGY MADRAS

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Data-Driven Optimization of Hospitality Operations:

A Case Study of Hotel SR Paark, Kavali

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Proposal for the Project on Business Data Management

Contents

1 **Executive Summary and Title** **4**

2 **Organisation Background** **4**

3 **Problem Statement** **5**

3.1 Problem statement 1 5

3.2 Problem statement 2 5

3.3 Problem statement 3 5

4 **Background of the Problem** **5**

5 **Problem Solving Approach** **6**

6 **Expected Timeline** **7**

7 **Expected Outcome** **8**

Declaration Statement

I am working on a Project Title "**Data-Driven Optimization of Hospitality Operations: A Case Study of Hotel SR Paark, Kavali**". I extend my appreciation to **Hotel SR Paark**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate

Name: Mannepalli Bala Praharsha

Date: 12-10-2025

Data-Driven Optimization of Hospitality Operations:

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1. Executive Summary

This research project is currently concerned with Hotel SR Paark, which is a hospitality business that is located close to the APSRTC bus stand in Kavali, Andhra Pradesh. The hotel mainly uses direct-customer strategy, which includes accommodation, a restaurant, and banquet services to a range of customers, including travelers, domestic families, and company clientele.

Key operational issues that were realized include changes in the occupancy of both guest rooms and banquet halls. Such fluctuations tend to create either underutilization or cases of overbooking. The mechanism of pricing is fixed and does not capture real time dynamics of demand. In addition, manual reservation of events initiates clashing of events, lost revenue opportunities and thus reduced efficiency and profitability.

In order to deal with these questions, the research will compile the primary information about the occupancy of rooms, event booking, and food and beverage sales within a set period of time. Regularities, the peak periods and bottlenecks in the operations will be identified using descriptive statistical analysis, trend analysis and graphical visualization.

The recommendations below underline the implementation of the dynamic pricing models, optimal room/event venue allocation, and simplified booking procedures. Such interventions are supposed to optimize the resource allocation and reduce inefficiencies in the scheduling.

The expected results are the creation of actionable insights that may help Hotel SR Paark to optimize capacity utilization, manage revenues better, and optimize the performance, and support the profitability and data-driven strategic planning to grow in the future.

2. Organization Background

Hotel SR Paark is a modern hospitality hotel located next to the APSRTC bus stand in Kavali, Andhra Pradesh and was started in 2024 by G.V. Narayana, M. Prasad and J. Bala Vardhan Reddy. The facility has comfortable accommodation and multi-purpose event space that serves both travelers, local families, and corporate customers. The guests will have the option of single suites and twenty-two deluxe rooms and also there is a banquet hall with dining services used during social as well as business meetings. The staff operations include a manager that oversees the general operations, a receptionist, 7 housekeeping staffs, and one security watchman. This team has a role of facilitating smooth day-to-day running. Hotel SR Paark targets mostly travelers, local families and event attendees with a focus on the guest satisfaction. It has a

strategic location that draws transient and local customers in the bus stand. The institution aims to provide comfortable, efficient, and friendly service as it strives to make enhancements in its service quality by making decisions based on data to promote sustainable growth.

3. Problem Statement

3.1 Occupancy and Booking Optimization:

- Since the room and banquet occupancy of SR Paark fluctuates, it leads to underutilization in weekdays and overbooking in high seasons.
- The current study is aimed at examining the trends in booking in detail and developing optimal strategies to be allocated.

3.2 Dynamic Pricing as a revenue maximization strategy:

- The current room and event space pricing strategies are mostly static and do not reflect demand variability.
- The paper will aim at suggesting the dynamic pricing models that can maximize revenue collection.

3.3 Event Booking Management:

- The current manual system of administering the banquet and event bookings creates issues of scheduling and missed opportunities.
- This project will involve a systematic review of the trends in bookings and will make substantive recommendations to improve the process of managing the events.

4. Background of the Problem

Hotel SR Paark is a small-scale hospitality business located in a tier-2 urban setting, which is faced with various operational challenges that undermine efficiency as well as profitability. The major problem relates to the changes in occupancy of its rooms and banquet halls, leading to underuse during the non-peak weekdays and overbooking during the high demand periods like festivals, weddings or corporate events. The causes of this fluctuation can be explained by unpredictable demand forecasting, lack of structured data and the use of ad hoc observational data in making decisions.

Additionally, the hotel does not incorporate seasonal changes in demand, special events, or market trends in its pricing strategy of both rooms and event space, thus curtailing its revenue earning potential, especially during high demand seasons. This shortcoming is also compounded by the fact that manual event booking processes are most likely to suffer scheduling clashes, duplicate bookings and lost business opportunities.

On the internal side, the business is also limited by the limited staff, the lack of automated tools, and a shortage of data regarding the occupancy and booking trends. On the outside, the volatile flows of tourists, local events, and competition in the market environment factors are also pertinent.

To address these issues, it becomes necessary to enumerate primary information pertaining to bookings, occupancy rates, and event dates, which are then subjected to analytical methods to reveal patterns, optimal price systems and smooth running of the event management systems.

5. Problem-Solving Approach

5.1 Methods and Justification:

The given study will be based on a set, data-driven approach that will help to mitigate operational inefficiencies at the hotel. The methodology is a combination of qualitative and quantitative methods to achieve the methodological and business relevance. These structured interviews with the management and employees will help in identifying problems in the processes at the level of the process of the organization and observing day-to-day running of the operations will serve to validate these observations. The following analysis of primary data on operational processes will indicate the trends in occupancy, bookings, and prices.

5.2 Data Collection:

The primary data were gathered within four weeks on the basis of reservation books, event registers, and point-of-sale books. Interest variables included type of room, dates of booking, hotel occupancy, details of event, booking channel, cancellations and prices. Data accuracy was guaranteed with the help of standardized templates, but the qualitative data about booking patterns and bottlenecks of the operation were also observed and documented.

5.3 Analysis Tools and Techniques:

The cleaning and preprocessing of data will occur in Microsoft Excel to address the issues of missing data, duplicates, and formatting issues. Following analytical processes, such as trend analysis, a correlation assessment, and scenario simulations, will be performed with the help of Python libraries, i.e., pandas, NumPy, matplotlib, and seaborn. Dashboards applied to Power BI or Tableau will be useful in ensuring the visualisation of key performance indicators (KPIs) like occupancy rate, Revenue per Available Room (RevPAR), event usage, and cancellations, thus producing actionable insights.

5.4 Justification of Approach:

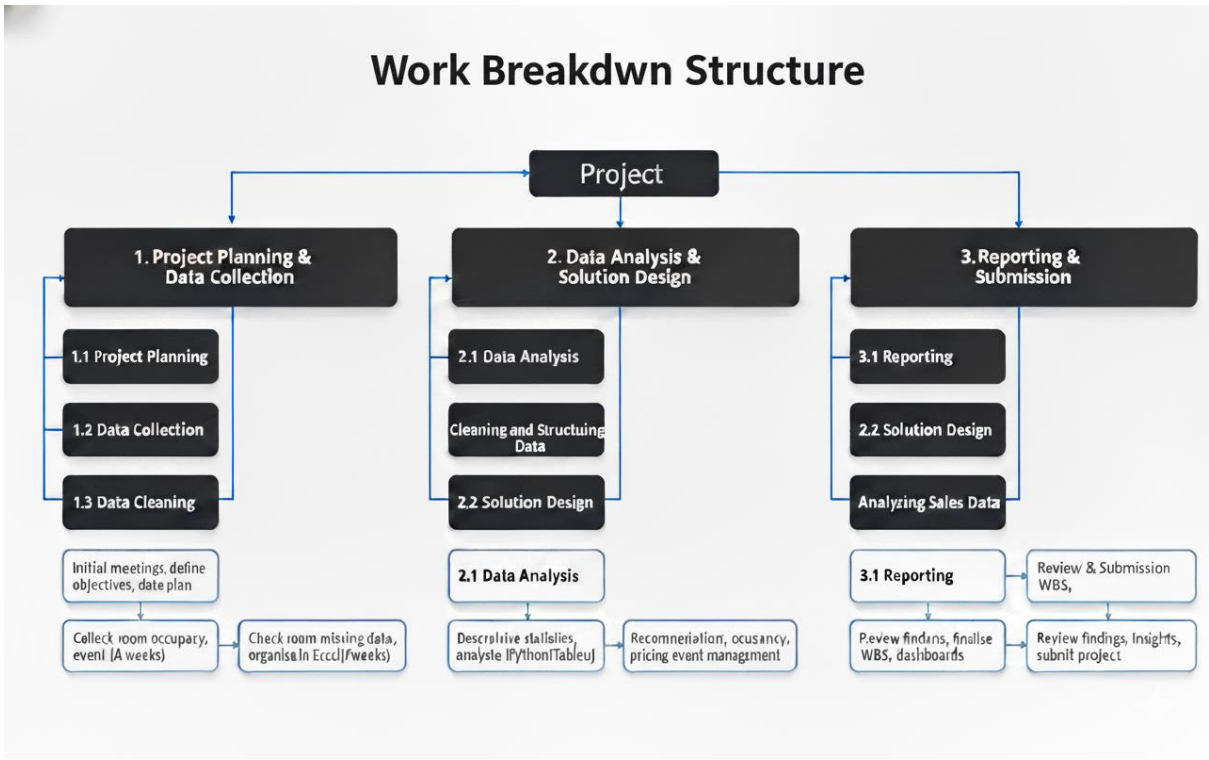
The methodology guarantees evidence-based recommendations since it incorporates primary data gathering, organized analytics, and visualization. Quantitative analysis helps to identify quantifiable inefficiencies, and interviews and observations are applicable to support operational and business relevance. This combined approach fills the gap between raw data and managerial interpretation, thus ensuring actionable and context-specific insights which inform decision-making.

5.5 Ethics and Data Privacy:

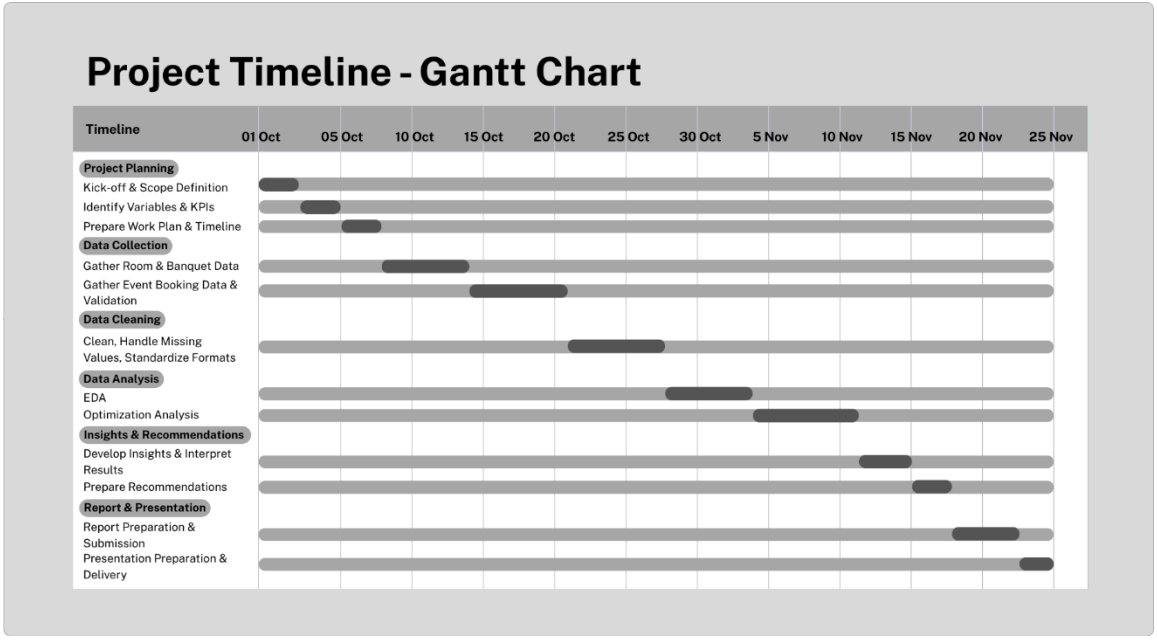
The project will be based on high levels of ethical research. Primary organizational data will not be shared with anybody and will be utilized solely in the academic analysis. Customer and employee information will not be released personally identifiable information. Data will be anonymized and will be stored securely and reported in aggregate. The collection of data will be done under the consent of the management to uphold ethical needs.

6. Expected Timeline

6.1. Work Breakdown Structure



6.2. Gantt Chart



6. Expected Outcome

The key goal of this project is to provide evidence-based information that will help Hotel SR Paark to improve the efficiency of operations, income generation, and customer satisfaction. The examination of the rooms and banquets attendance will provide the establishment with an accurate insight into high and low seasons and, as a result, allow to allocate rooms and events areas in the most optimal way. Such strategy will reduce underutilization on weekdays and prevent overbooking on high-demand periods.

The dynamic pricing recommendations will give the hotel the ability to adjust room and banquet rates based on the demand and hence optimize its revenue without effect on the satisfaction of the guests. Event-booking trend analysis will also provide more insight into the common scheduling issues and inefficiencies and enable the management to optimize booking operations and eliminate the lost opportunities.

Further, the project will separate critical determinants which impact the occupancy and revenue, including the type of customers, the booking methods and seasonal patterns. The results will provide practical recommendations that SR Paark may use to streamline its planning, resource allocation and operations decision making.

In general, the expected result will ensure that this hotel will be able to use empirical data to implement informed decisions, increase profitability, and provide a higher level of experience to its guests and event clients.